Multichannel Interconnection Decomposition for Hyperspectral LiDAR Waveforms Detected From Over 500 m

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***Abstract*— The full-waveform hyperspectral light detection and ranging (FWHSL) data have been widely used in surface topography, vegetation detection, and 3-D urban terrain modeling, capable of revealing the spatial distribution of a target and more detailed spectral information in the vertical direction. However, the echo signals of a target would significantly vary between different spectral channels due to the reflectance characteristics and the uneven energy distribution of supercontinuum laser source. Especially, band channels with weak reflectance over a long distance would affect the extraction accuracy of waveform parameters, which are essential for retrieving the spatial and spectral information of targets. This article proposes a multichannel interconnection decomposition method to improve the extraction accuracy of distance and spectral information at each pulse using hyperspectral waveform data. Two experiments were conducted to verify the performance of long-distance detection of targets using FWHSL. The first experiment detected a standard whiteboard, a green leaf, and a yellow leaf at roughly 518 m. Results demonstrated a considerable improvement in ranging precision and spectral detection using the proposed method compared with using the optimal channel with the best data quality. The second experiment simultaneously detected two adjacent targets at a distance of approximately 518 m. Results presented clear superiority of adding waveform channels in terms of discovering overlapping components and retrieving accurate waveform parameters. The success rate of extracting two targets 60 cm apart was greatly increased from 47% to 73% through the multichannel interconnection waveform decomposition (MIWD) method.**

***Index Terms*— Full-waveform light detection and ranging (LiDAR), hyperspectral LiDAR, ranging, spectral detection, waveform decomposition.**

I. Introduction

In THE past decade, full-waveform light detection and ranging (LiDAR) has been increasingly applied to surface topography, land cover classification, vegetation detection, and 3-D urban terrain modeling [1]–[5]. Previous researches have demonstrated the great advantages of the whole backscattered signals recorded by full-waveform LiDAR systems compared to discrete return systems [6], [7]. The abundant information contained in the waveform data helps improve the range accuracy [8], [9] and retrieve the targets within the footprint in detail related to weak and overlapping pulses [6], [10], [11]. In addition, the reflection and surface geometric characteristics of targets can be retrieved from waveforms [12]–[15]. Therefore, the full-waveform LiDAR systems exhibit outstanding advantages in various applications.

With the development of supercontinuum laser source and highly sensitive detection technology, considerable efforts have recently been exerted for full-waveform multi/hyperspectral LiDAR (FWMSL/FWHSL) systems design and evaluation [16]–[19]. These FWHSL systems can record spatial distribution and hyperspectral information simultaneously in the vertical direction with multiple echoes from different targets within the laser footprint [20]. Previous studies mainly focused on the spectral resolution improvement by increasing receiving spectral channels to enhance the spectral detection capability, which plays an important role in obtaining vertical spectral and biochemical distribution of vegetation [18], [21], [22], as well as land cover classification [23]–[25]. Despite this remarkable progress, few studies have made a breakthrough in long-distance detection. Almost all the related hyperspectral LiDAR experiments were carried out in laboratory circumstances from a few meters to a maximum effective range of roughly 20 m [26]–[28]. Short-distance detection limits further development of hyperspectral LiDAR, especially for verifying its large-scale detection capabilities. Thus, long-distance ($>$500 m, which is a suitable distance for the airborne platform) detection is crucial to realize the potential application ability of this new LiDAR technology [29]–[32].

However, with the detection distance increasing from nearly 10 m to several hundred meters, the difficulty of multichannel waveform data processing would increase significantly, when considering the influence of outdoor environment. In long-distance scenarios, the echo signal decreases quadratically with the distance. Meanwhile, the echo signal fluctuation between different spectral channels would become obvious due to the reflectance characteristics of targets and the uneven energy distribution of supercontinuum laser sources [28]. Particularly, the channels with weak reflectance over long

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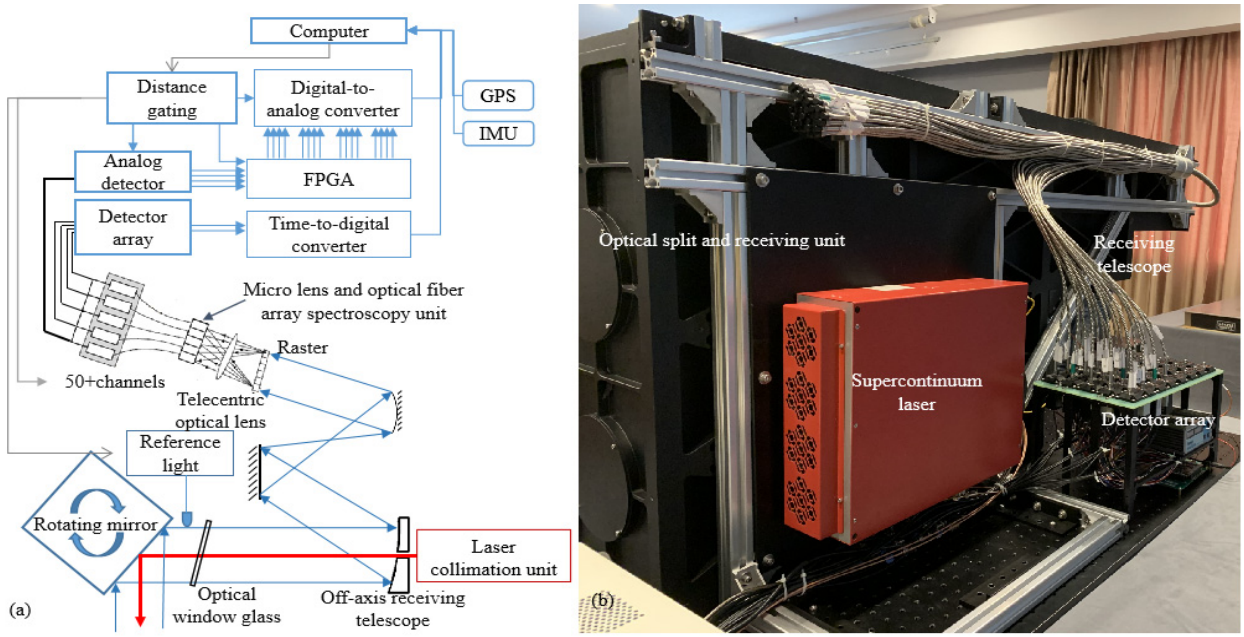


Fig. 1. (a) Architecture of an FWHSL system and (b) FWHSL prototype.

distances will hinder the extraction of waveform parameters, namely, amplitude, central location, and full width at half maximum (FWHM), which are crucial for target property retrieval [19], [33], [34]. Moreover, given that detecting the initial waveform peak from the waveform data is difficult, the relatively poor signal quality will be detrimental to the separation between overlapping echo components from adjacent targets [11], [35]. Nonetheless, the abundant information from multichannel waveforms could provide great advantages in discovering potential waveform components, including the weak and overlapping echo components, thus help to improve the discrimination of the adjacent targets, which are hard to separate from the traditional single-wavelength LiDAR [19], [36]. In addition, the raw data quality of each receiving channel could be improved by multichannel waveform accumulation, thereby obtaining relatively high single-pulse echo positioning precision. Therefore, the further development of FWHSL should dedicate to utilize the existing multichannel waveform advantages to improve the accuracy of spatial and spectral information extraction, and ultimately achieve long-distance detection, which makes this new technology practical in the large-scale detection.

This article introduces a prototype of an airborne FWHSL system designed for vegetation applications that can record backscatters in over 50 spectral channels from 400 to 900 nm, with a spectral sampling interval of roughly 10 nm [37]. Furthermore, we propose a multichannel interconnection waveform decomposition (MIWD) method to explore the potential capability of long-distance ($>$500 m) detection using the FWHSL data. The main objects of this article can be summarized as follows.

1) This article proposes an MIWD method that can retrieve the waveform parameters of echo components at multiple spectral channels.
2) To utilize the waveform data of numerous spectral channels, we propose a dynamic optimization strategy of multichannel waveform accumulation. The channels participating in the waveform accumulation and the corresponding constraint weights are dynamically adjusted through waveform quality analysis.
3) To verify the ability of the proposed method to extract weak or overlapping signals in long-distance detection, we design a set of experiments at a distance of roughly 518 m. The ranging accuracy, multitarget resolution, and spectral detection capability are evaluated on typical targets.

The remainder of this article is organized as follows. Section II describes the FWHSL system and experimental design. Section III explains the proposed method in detail. Section IV demonstrates the method by applying it to real datasets, and Section V discusses the experimental results. Finally, Section VI outlines the conclusion and potential future research directions.

II. Materials

A. FWHSL System

The architecture of the FWHSL system is depicted in Fig. 1. The system was established in the application of target identification and observation, especially in the monitoring of growth and nutritional status of vegetation. The system transmits 4-ns pulses at a repetition rate of 63 kHz and peak power of 20 kW produced by a supercontinuum laser. The broadband white light emitted by the laser is collimated (diverging angle less than 0.5 mrad), when it passes through the refracting collimator. The effective aperture of the receiving telescope is designed as 170 mm, which not only meets the need for a return signal detection but also reduces the size and weight of

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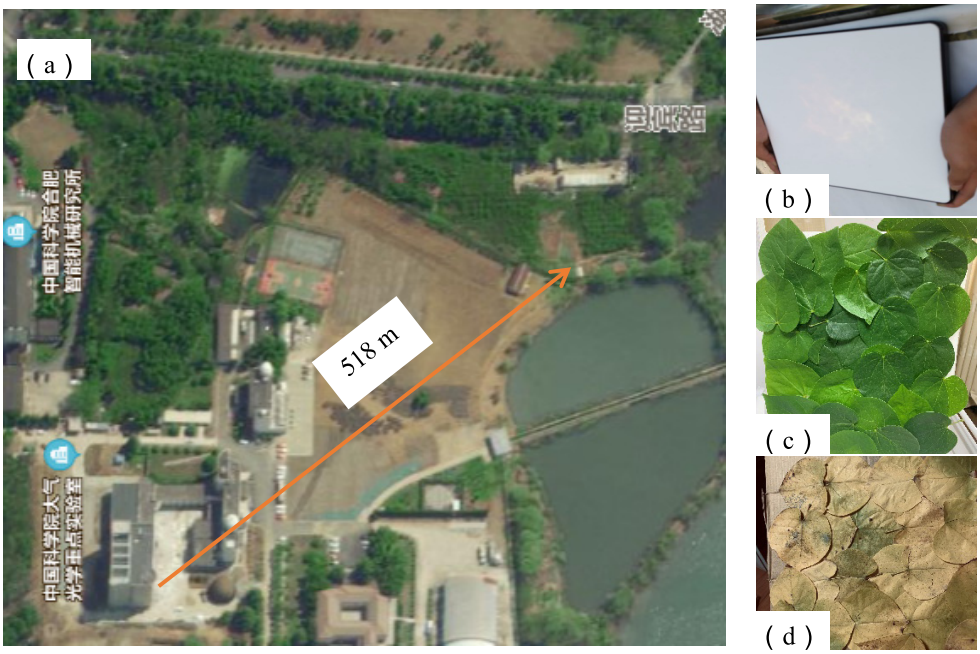


Fig. 2. (a) Experimental area, around 518 m away from the FWHSL system, and detection targets. (b) Standard whiteboard, (c) green leaves, and (d) yellow leaves.

the telescope. The system adopts a relatively small receiving field of view and a complicated optical path, so as to minimize the influence of background light and improve the signal-to-noise ratio (SNR) of the echo signal from a long distance. The transmitting and returned waveforms are digitalized in 12-bit analog-to-digital converter at a sampling rate of 2 GHz.

The spectral range of output laser is designed as 400–900 nm. The scattered laser pulse from the target is filtered and resolved by an optical grating system to contain 56 spectral channels with a spectral resolution of nearly 10 nm. The spectral channels can be flexibly selected for simultaneous detection on the basis of the characteristics of the detected targets. In this work, 37 visible channels at the wavelengths shorter than the red edge domain of vegetation (436–776 nm) and three near-infrared channels (785–804 nm) were selected, which facilitate the construction of various vegetation indices. The central wavelengths for the selected spectral channels are listed in Table I. The spectrally separated light is converted to analog voltages using photomultiplier tube (PMT) detector array.

TABLE I
CENTRAL WAVELENGTHS FOR THE 40 SPECTRAL CHANNELS IN THE HYPERSPECTRAL LiDAR

Channel	CH1	CH2	CH3	CH4	CH5
Wavelength(nm)	436	445.6	461.6	470.4	479.4
Channel	CH6	CH7	CH8	CH9	CH10
Wavelength(nm)	488.2	497.2	506.4	516.2	525
Channel	CH11	CH12	CH13	CH14	CH15
Wavelength(nm)	534.8	544	553	563	571.8
Channel	CH16	CH17	CH18	CH19	CH20
Wavelength(nm)	580.6	589.6	598.8	608	617.6
Channel	CH21	CH22	CH23	CH24	CH25
Wavelength(nm)	626.8	635.6	645.8	654.6	664.8
Channel	CH26	CH27	CH28	CH29	CH30
Wavelength(nm)	674.6	683.6	692	702	711.4
Channel	CH31	CH32	CH33	CH34	CH35
Wavelength(nm)	720.4	730.8	738.4	748.2	758.2
Channel	CH36	CH37	CH38	CH39	CH40
Wavelength(nm)	766	776	785	796.4	804

Chi: the label of the spectral channel

B. Experimental Design

To verify the performance of the proposed method using FWHSL under strong ambient light in long-distance detection, the detection experiments are designed into two steps: the first step verifies the ability of the proposed method in ranging and spectral retrieval; the other step evaluates the ability of the method to obtain the spatial distribution of targets along the laser path with high resolution. Fig. 2 shows the experimental area and detection targets. Given that FWHSL has great advantages in vegetation remote sensing applications, the representative targets, including green and yellow leaves with different health conditions and a standard whiteboard as a reference target, are detected at roughly 518 m away from the system.

Particularly, in the first step, the three representative targets were detected successively in the dome building [Fig. 2(a)]. To avoid the interference of sub-footprint, we visually ensured that the entire laser footprint was illuminated onto the target surface vertically. In the next step, two standard whiteboards in the dome building were detected simultaneously, as shown

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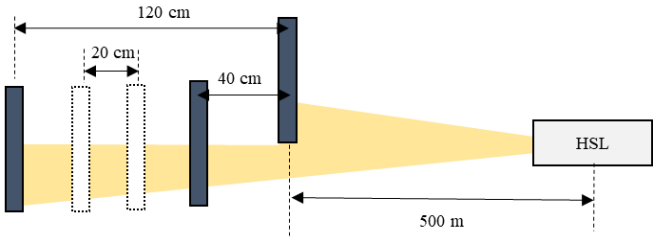


Fig. 3. Experimental setup used to obtain the laser echoes scattered from two close targets with varying intervals.

in Fig. 3. The laser footprint partially hits the two standard whiteboards, and the superposition of the laser pulses scattered from these two planes was collected. The distance between these two targets was changed from 40 to 120 cm with a step size of 20 cm by moving the second plane away from the first plane. For each target in the first step and every two targets in the second step, 1000 repeated measurements were performed, and the corresponding waveform data were collected to subsequently evaluate the performance of the proposed method.

III. Multichannel Interconnection Waveform Decomposition Method

The echo signal usually exhibits significant variations between different spectral channels during the long-distance detection due to the target reflectance and the uneven energy distribution of supercontinuum laser source. The signal characteristics of the long-distance detection would greatly affect the extraction accuracy of waveform parameters and the performance of waveform ranging especially for weak or overlapping signals. Moreover, the pulse-echo arrival time of each spectral channel is not strictly synchronized which obscures the retrieval of geometric information. Therefore, considering the spatial correlation and shape characteristics of the echo waveforms in each spectral channel, an MIWD method is proposed to improve the extraction accuracy of waveform components from multiple targets. Generally, the MIWD method is performed in four steps, including waveform preprocessing, multichannel weighted waveform accumulation (MWWA), hyperspectral waveform parameter extraction (HWPE), and accuracy assessment, as shown in Fig. 4.

A. Preprocessing

Given that no signal echo appeared before the transmitted pulse, the beginning part of the waveforms is considered the stable background noise, which is used to estimate the noise level. Then, a noise threshold is set as three times the standard deviation over the mean background noise to distinguish signal from noise since the background noise is considered to satisfy a normal distribution, expressed as

$$\text{thr}_{\text{noise}} = \mu_{\text{noise}} + 3\sigma_{\text{noise}} \tag{1}$$

where μ_{noise} and σ_{noise} are the mean value and standard deviation of the beginning part of the waveforms. $\text{thr}_{\text{noise}}$ is the noise threshold to distinguish signal from noise. The waveform signal is then smoothed with a Gaussian filter whose width is set as the width of the transmitted pulse to remove the interference of background noise.

Multichannel echo signals from the same observation target may have different pulse-echo arrival times, which complicate the retrieval of the target spatial distribution. The difference in the pulse-echo arrival time is affected by many factors, including the propagation time difference in the fixed-length optical path of multiple spectral channels inside the instrument, the response of the multichannel photodetectors, and analog-to-digital conversion. In order to enhance the long-distance detection capability, the optical path of the FWHSL system is complicated and long as described in Section II-A, which aggravates the influence of the propagation time difference of each spectral channel. Therefore, the error of the recorded time from the transmitted pulse and target echo in each spectral channel can be divided into two parts and expressed as

$$\varepsilon_{t,\lambda} = \varepsilon_\text{def}_{t,\lambda} + \varepsilon_\text{indef}_{t,\lambda} \tag{2}$$

$$\varepsilon_{r,\lambda} = \varepsilon_\text{def}_{r,\lambda} + \varepsilon_\text{indef}_{r,\lambda} \tag{3}$$

where $\varepsilon_{t,\lambda}$ and $\varepsilon_{r,\lambda}$ represent the error of the recorded time from the transmitted pulse and target echo in each spectral channel, respectively. $\varepsilon_\text{def}_{t,\lambda}$ and $\varepsilon_\text{def}_{r,\lambda}$ represent the corresponding definite part, related to the optical path setting of the FWHSL system. $\varepsilon_\text{indef}_{t,\lambda}$ and $\varepsilon_\text{indef}_{r,\lambda}$ represent the corresponding indefinite part, related to photodetection and analog-to-digital conversion.

The propagation time difference in the fixed-length optical path inside the instrument is calculated on the basis of the refractive index of the optical fiber, as in (4) and (5). Given that the transmitted and target echo go through the same photodetector and analog-to-digital channel, the corresponding indefinite part $\varepsilon_\text{indef}_{t,\lambda}$ and $\varepsilon_\text{indef}_{r,\lambda}$ can be regarded as identical. Therefore, the flight time of the target echo is calculated with the time of the transmitted pulse as the reference time as in (6)–(8) to reduce the random errors

$$\varepsilon_\text{def}_{t,\lambda} = \text{length}_t / c^* n_{t,\lambda} \tag{4}$$

$$\varepsilon_\text{def}_{r,\lambda} = \text{length}_r / c^* n_{r,\lambda} \tag{5}$$

where c represents the speed of light in vacuum. $n_{t,\lambda}$ and $n_{r,\lambda}$ represent the refractive indices of the optical fibers, e.g., fused silica. length_t and length_r represent the length of the optical fibers

$$\begin{aligned} t &= (\text{time}_{r,\lambda} - \varepsilon_{r,\lambda}) - (\text{time}_{t,\lambda} - \varepsilon_{t,\lambda}) \\ &= (\text{time}_{r,\lambda} + \varepsilon_\text{indef}_{r,\lambda} + \varepsilon_def_{r,\lambda} - \varepsilon_\text{def}_{t,\lambda}) \\ &\quad -(\text{time}_{t,\lambda} + \varepsilon_\text{indef}_{t,\lambda}) \end{aligned} \tag{6}$$

$$C_{t,\lambda} = \varepsilon_\text{indef}_{t,\lambda} \tag{7}$$

$$C_{r,\lambda} = \varepsilon_\text{def}_{t,\lambda} - \varepsilon_\text{def}_{r,\lambda} + \varepsilon_\text{indef}_{r,\lambda} \tag{8}$$

where t represents the flight time of the target echo that is calculated with the time of the transmitted pulse as the reference time. The echo waveform of each channel is then unified in time domain after multichannel time synchronization calibration, so that the position distribution of all channels can be integrated into the subsequent data processing.

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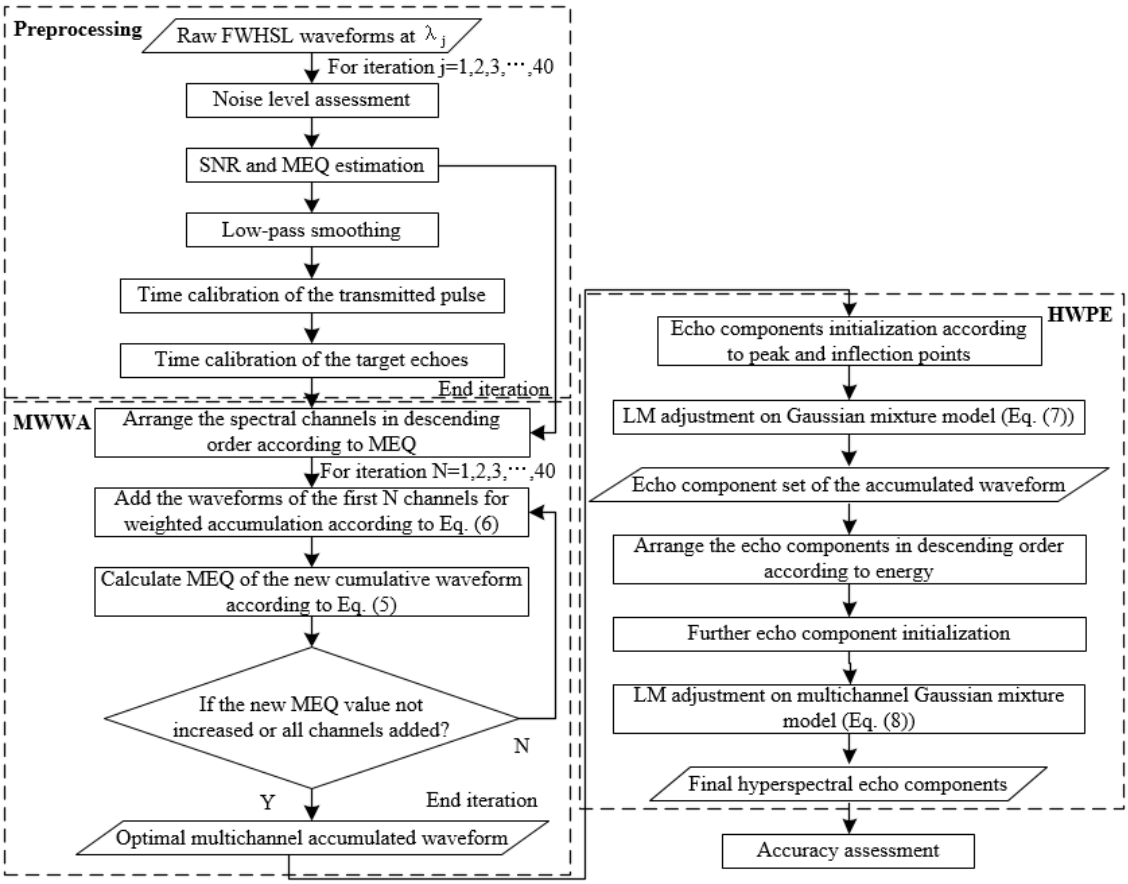


Fig. 4. Flowchart of FWHSL processing, comprising waveform preprocessing, MWWA, HWPE, and accuracy assessment.

B. Multichannel Weighted Waveform Accumulation

Although the return time of the hyperspectral LiDAR waveforms has been completely synchronized in the preprocessing so far, the echo signals of different receiving bands exhibit obvious differences in strength due to the different reflection characteristics of targets. The extraction accuracy of waveform components would be seriously affected, when based on a single channel especially for low reflectivity bands, and some weak signals with low response would even be lost. However, the determination of the number and approximate location of all targets in the laser footprint is extremely important for retrieving spatial and spectral information.

To detect potential targets corresponding to weak echoes and obtain more accurate location distribution information of the targets, the waveforms of each channel are accumulated to enhance weak signals and improve the overall data quality, which is similar to multipulse accumulation in single channel. Unlike the latter, the quality of the waveform data involved in the accumulation process varies in multiple channels. In case the channels with high noise levels may bring side effects in the process of multichannel waveform accumulation, a dynamic optimization strategy is designed that can adaptively adjust the participating spectral channels and set the corresponding constraint weights on the basis of the data quality of each channel.

The data quality of the echo waveform is usually expressed in SNR, which is calculated using the ratio between the maximum value of the returned signal and the standard deviation of the background noise as in (9). However, SNR can only evaluate the maximum value in the echo waveform, which is not suitable for echo waveforms containing multiple waveform components. Therefore, a new waveform quality evaluation index, multiecho quality (MEQ), as in (10) is proposed

$$\mathrm{SNR} = 20\log_{10}(S_{\max}/N_{\mathrm{std}}) \tag{9}$$

$$\mathrm{MEQ} = 20\log_{10}\left(\frac{\sum_{x=1}^{M} f(x)}{N_{\mathrm{std}}}\right) \tag{10}$$

where $S_{\max}$ is the maximum value of each echo waveform. $f(x)$ represents the multiecho waveform, and M represents the length of the waveform. The value of MEQ is positively correlated with the quality of the waveform.

If only one target in the laser footprint exists, the spectral channels are added in the accumulation in descending order on the basis of the SNR or MEQ values; otherwise, the channels are added in descending order on the basis of the MEQ values by default. To maintain the scale of the waveform intensity and further suppress the influence of background noise, two forms of weighted accumulations are proposed: 1) the weight of each channel is the same and 2) the weight of each channel is inversely proportional to the corresponding background noise level as in (11) and (12). The first weighted accumulation

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method is suitable for the situation where the noise level of each channel is similar and has higher computing efficiency; the latter is dedicated to suppressing the influence of high noise level channels on the accumulated waveform

$$w_j = \begin{cases} 1/N \\ \dfrac{1/\sigma_{\text{noise},j}}{\sum_{i=1}^{N} 1/\sigma_{\text{noise},i}}, \end{cases} \quad j = 1, 2, \ldots, N \tag{11}$$

$$f_a(x) = \sum_{j=1}^{N} f_j(x)^* w_j \tag{12}$$

where N represents the number of spectral channels involved in the accumulation, $f_j(x)$ represents the waveform data of the jth channels, and $f_a(x)$ represents the accumulated waveform. w_j represents the weight of the corresponding waveform. During the multichannel waveform accumulation, at each time, the waveform data of a spectral channel is added, the data quality (SNR or MEQ) of the new weighted waveform $f_a(x)$ will be calculated again on the basis of (12). This process will stop until the new data quality is not improved or all channels participate in the accumulation.

C. Hyperspectral Waveform Parameter Extraction

Benefiting from the weighted accumulation of multichannel waveforms, the potential echo components of the accumulated waveform are enhanced, thereby simplifying the retrieval of the target position distribution. Given that the accumulated waveform contains all the enhanced echo components, the position distribution of targets can be retrieved using the traditional single-wavelength waveform decomposition (SWWD) method, Gaussian waveform decomposition as given in the following equation [12], [38]:

$$f(t) = \sum_{i=1}^{K} a_i \exp\left(-\frac{(t-c_i)^2}{F_i^2/(4\ln 2)}\right) \tag{13}$$

where $f(t)$ represents the accumulated waveform data as a function of time t, the same as $f_a(x)$ in (12). K represents the number of waveform components, including a transmitted pulse and several target echoes. a_i, c_i, and F_i represent the waveform parameters, i.e., amplitude, central location, and FWHM of the ith waveform components, respectively. Among the waveform parameters, the central location c_i, also called the return time of a waveform component, is related to the position distribution of all targets within the footprint. The waveform components are first initialized on the basis of the peak and inflection points, and then the optimal solution of each waveform parameter is obtained through iterative optimization, i.e., the Levenberg–Marquardt (LM) method [39]. The potential echo components are enhanced through accumulation, making the initialization close to the true value. Therefore, the accumulated waveform component set is obtained, and the position distribution of the targets within the laser footprint will have a fine retrieval.

In addition to the position distribution of the echo components, other waveform parameters, i.e., amplitude and FWHM, of each spectral channel must be further extracted to retrieve the spectral and surface geometry information of the targets. The hyperspectral echo waveforms can be fit using the multichannel Gaussian mixture model as in (14) on the basis of our previous study [19]. Given that the spatial distribution of the waveform components has been accurately determined previously, the center location of each echo component is fixed in the multichannel Gaussian mixture model. Therefore, the algorithm can avoid falling into local optimization in the iterative optimization process, and further, the weak echoes or even overlapping echoes are expected to be accurately extracted with high robustness

$$f_{\lambda_j}(t) = \sum_{i=1}^{K} a_{i,\lambda_j} \exp\left(-\frac{(t-c_i)^2}{F_{i,\lambda_j}^2/(4\ln 2)}\right) \tag{14}$$

where $f_{\lambda_j}(t)$ represents the waveform data of the jth spectral channel. a_{i,λ_j}, c_i, and F_{i,λ_j} represent the waveform parameters of the jth spectral channel, respectively. The new waveform parameters are initialized on the basis of the accumulated waveform component set $\{a_i, c_i, F_i\}_{\text{accu}}$. The amplitude is initialized as the intensity of the point in the location c_i. The FWHMs are initialized as the values in $\{a_i, c_i, F_i\}_{\text{accu}}$ given that the echo width of each channel is similar. After initialization, the LM optimization is deployed once more in the parameter adjustment to adjust the echo components to approximate the raw waveform data.

Furthermore, as the transmitted and target echo waveforms are simultaneously recorded by the FWHSL system, the relative power rP_{i,λ_j} of the ith echo component is defined as the ratio of received and transmitted energy to eliminate the influence of laser energy instability [18]. Then, a normalized reflectance R_{i,λ_j} is calculated as the ratio between the relative powers of a leaf and the standard whiteboard as given in the following equations:

$$P = \int a_{i,\lambda_j} \exp\left(-\frac{(t-c_i)^2}{F_{i,\lambda_j}^2/(4\ln 2)}\right) dt \tag{15}$$

$$rP_{i,\lambda_j} = \frac{P_{i,\lambda_j}}{P_{t,\lambda_j}} \tag{16}$$

$$R_{i,\lambda_j} = \frac{rP_{\text{leaf},i,\lambda_j}}{rP_{\text{reference},i,\lambda_j}} \tag{17}$$

where the power P of the waveform signal is expressed as function integral value. P_{t,λ_j} represents the power of transmitted pulse in the wavelength of λ_j. P_{i,λ_j} represents the power of the ith echo component in the wavelength of λ_j. $rP_{\text{leaf},i,\lambda_j}$ and $rP_{\text{reference},i,\lambda_j}$ represent the relative powers from a leaf and the reference target, respectively. In practice, the echo waveform at each wavelength λ_j is collected using a 99% standard reflectance panel as a reference target. The normalized reflectance of the measured target can be derived once the relative power of the reference target is determined in the laboratory.

D. Accuracy Assessment

The MIWD method retrieves the hyperspectral waveform parameters, which are related to the position distribution and spectral reflectance characteristics of targets. To evaluate the performance of the proposed method, the standard deviations

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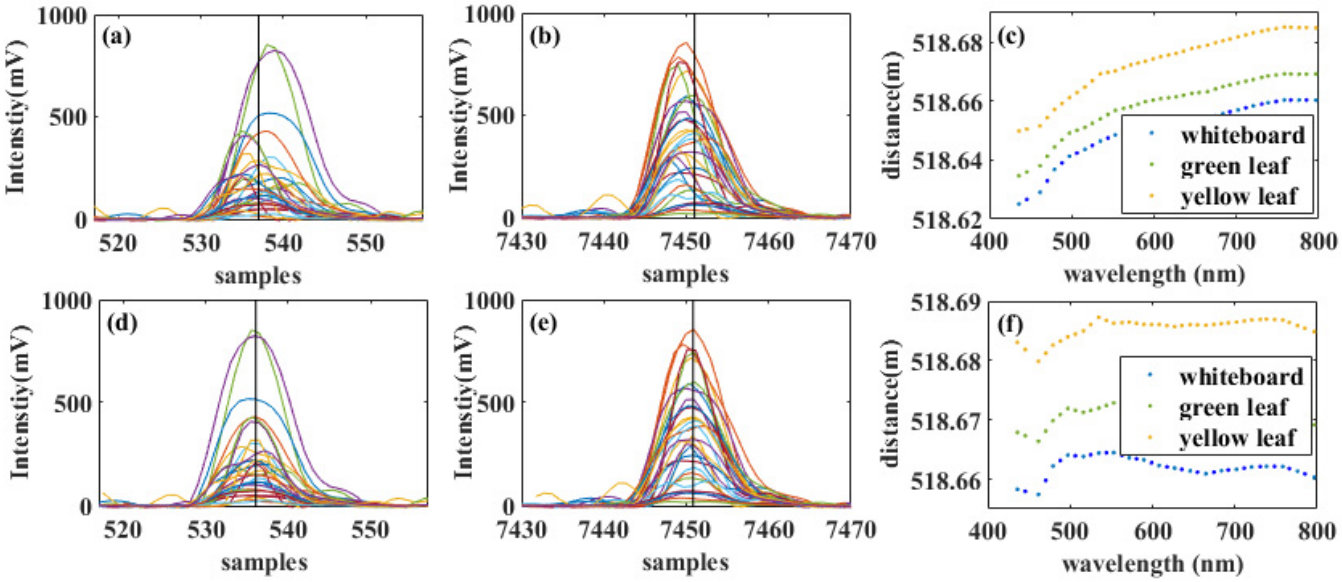


Fig. 5. Waveform display and ranging results before and after multichannel time synchronization calibration. (a) and (b) Raw waveforms of transmitted pulse and target echo at each spectral channel, while (d) and (e) results after multichannel time synchronization calibration. The colored solid splines represent the signals at each channel after removing the background noise. The vertical lines indicate the peak positions of the longest wavelength channel (804 nm). (c) and (f) Ranging results at different spectral channels before and after calibration. The blue, red, and yellow scattered spots represent the ranging results of the whiteboard, the green leaf, and the yellow leaf, respectively.

of the distance and spectrum measurements are employed to assess the ranging and spectral accuracy. Moreover, the waveform decomposition accuracy is evaluated through coefficient of determination (R^2), root mean square error (RMSE), and relative neighbor distance error (RNDE), as shown in (18)–(20). R^2 and RMSE reflect the overall fitting accuracy between the fit waveform and the raw waveform [40], while the RNDE represents the relative error in the distance between adjacent targets [19]

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y}_i)^2} \tag{18}$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2} \tag{19}$$

$$\text{RNDE} = \frac{|\text{MND} - \text{TND}|}{\text{TND}} \tag{20}$$

where $\hat{y}_i$ is the estimated value of waveform data, y_i is the measured value of waveform data, and $\bar{y}$ is the average measured value of waveform data. True neighbor distance (TND) represents the true value of the distance between adjacent targets within a footprint and measured neighbor distance (MND) represents the measured value of the distance between the adjacent waveform components decomposed through a waveform decomposition method.

IV. Results

A. Calibration of the Time Synchronization at Different Spectral Channels

Fig. 5 displays the comparison results before and after the time synchronization calibration. Specifically, Fig. 5(a) and (b) presents the transmitted pulse and the target echo for each channel before calibration. As discussed in Section III-A, it can be seen from the peak positions of the recorded waveforms that there is an obvious time deviation of up to 3 ns between the spectral channels, which limits the combination of multiple channels for ranging and weak echo detection. Therefore, the traditional SWWD method as mentioned in Section III-C is deployed in the waveforms of each channel in order to analyze the ranging performance of the system without the synchronization error being calibrated, as shown in Fig. 5(c). The results show that, for each target, the ranging deviation between the spectral channels is as high as roughly 4 cm, which extremely hinders the extraction of mutually corresponding targets in each channel.

Moreover, after the synchronization calibration, the peak positions between multiple channels are almost the same as shown in Fig. 5(d)–(e), which means that the time asynchrony is well eliminated. In addition, Fig. 5(f) demonstrates the corresponding ranging results using SWWD, in order to analyze the ranging performance of each channel after calibration. Benefiting from the time synchronization calibration between multiple channels, the ranging deviation between channels is greatly reduced from 4 cm to roughly 1 cm. Furthermore, to quantify the influence of time synchronization calibration on ranging, the maximum ranging deviation between the spectral channels and the standard deviation of the measured values of multiple channels are listed in Table II.

Table II illustrates the significant improvement of the ranging deviation between channels through the time synchronization calibration. Specifically, the maximum ranging deviation between channels, that is, the difference between the maximum and minimum values, is reduced by roughly five times (before: 3.54, 3.46, and 3.53 cm, after: 0.71, 0.65, and 0.74 cm) through the time synchronization calibration. Moreover, the standard

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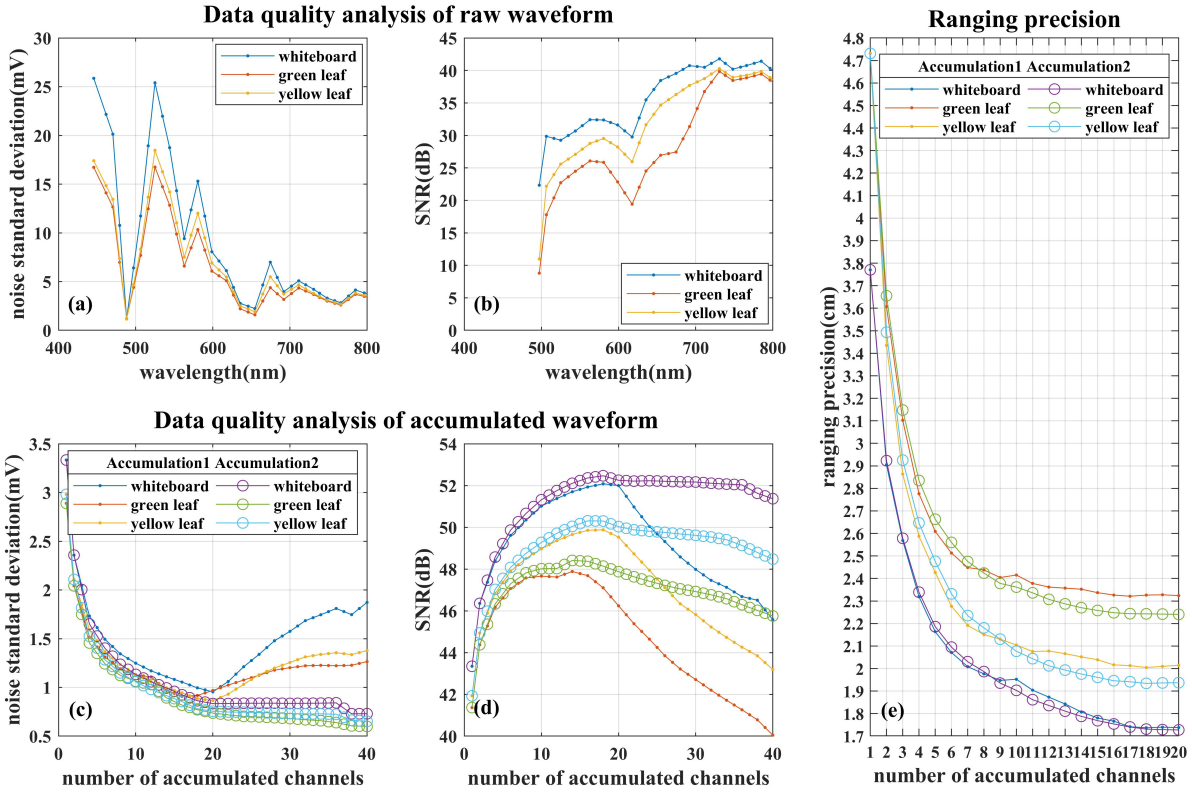


Fig. 6. Data quality analysis of the raw and accumulated waveforms and their ranging results. (a) and (b) Data quality analysis of the raw waveforms in terms of noise standard deviation, and SNR. (c) and (d) Data quality analysis of the two kinds of accumulated waveforms. (e) Results of ranging precision as the accumulated channels of two kinds of accumulated waveforms increase. The colored dotted lines correspond to the whiteboard, the green leaf, and the yellow leaf, respectively.

TABLE II
STATISTICAL RANGING RESULTS OF THE WHITEBOARD, THE GREEN LEAF, AND THE YELLOW LEAF, INCLUDING THE MAXIMUM RANGING DEVIATION BETWEEN THE SPECTRAL CHANNELS, AND THE STANDARD DEVIATION OF THE MEASURED VALUES OF MULTIPLE CHANNELS

Methods	Max-min value (cm)			Standard deviation (cm)		
	W	G	Y	W	G	Y
Before calibration	3.54	3.46	3.53	1.02	1.03	1.11
After calibration	0.71	0.65	0.74	0.18	0.16	0.18

W: whiteboard, G: green leaf, Y: yellow leaf.

deviation between channels is improved over five times from 1.02, 1.03, and 1.11 cm to 0.18, 0.16, and 0.18 cm. Thus, the time synchronization calibration can effectively improve the ambiguity of spatial information between channels, and help to combine and utilize the waveforms of multiple channels in subsequent data processing. This process is crucial for determining the same detected target for each channel, as long as the target has an obvious response in a certain channel, even if the echoes are weak, the corresponding echoes can be found in the other channels.

B. Multichannel Ranging and Spectral Information Retrieval

In order to explore the advantages of combining multichannel waveforms in the long-distance detection, the ranging precision and spectral acquisition capability are further evaluated after the time synchronization calibration between channels. Fig. 6 demonstrates the considerable improvement in data quality and ranging precision through the multichannel accumulation. The data quality analyses in Fig. 6(a) and (b) verify that the background noise level and SNR vary significantly in different spectral channels. Specifically, the detection channels in the long-wavelength region (500–800 nm) show a higher SNR and a lower noise level than those in the short wavelength region (400–500 nm), especially for the green leaf. Nonetheless, the short wavelength region also plays an important role in detecting biochemical characteristics of vegetation. Therefore, the corresponding poor data quality must be overcome to extract accurate waveform information. In the MIWD algorithm, the waveforms in the long-wavelength region can provide a reference position of each echo component, thus improving the extraction accuracy in the short wavelength region.

Given that each recorded waveform in the first experiment contains only one echo component, the waveform data are accumulated orderly on the basis of the SNR values to further improve the whole data quality. Fig. 6(c) and (d) displays the improvement of the data quality in terms of noise standard deviation and SNR as the number of the accumulated channels increases. Specifically, when the number of accumulated channels is less than 20, the background noise level decreases, and the SNR values increase rapidly as the number of accumulated channels increases; thereafter, the data quality is obviously

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TABLE III

STATISTICAL RESULTS OF WAVEFORMS FROM SINGLE CHANNEL (730 nm) AND TWO KINDS OF ACCUMULATED WAVEFORMS, INCLUDING NOISE STANDARD DEVIATION (N_{STD}), SNR, AND RANGING PRECISION (E_p)

Methods	Nstd (mV)			SNR (dB)			Ep (cm)		
	W	G	Y	W	G	Y	W	G	Y
Single channel (730 nm)	4.20	3.67	3.79	41.79	39.85	40.28	3.77	4.73	4.73
Accumulation1	0.95	0.91	0.86	52.08	47.88	49.88	1.74	2.32	2.00
Accumulation2	0.84	0.66	0.73	52.47	48.41	50.31	1.73	2.24	1.93

W: whiteboard, G: green leaf, Y: yellow leaf.

affected by the poor spectral channels. The multichannel accumulation process in the MIWD method includes equal-weighted accumulation and weighted accumulation based on the background noise level. The background noise level and SNR verify that the second accumulation method is more robust and less susceptible to the influence of the spectral channels with weak echoes compared with the first one. The results in Fig. 6(c) and (d) indicate that choosing 20 channels with the best signal quality can generate the optimal accumulated waveform. Thus, the change in ranging precision as the number of accumulated channels increases from 1 to 20 is presented in Fig. 6(e). Generally, the ranging precision is greatly improved with the addition of accumulated channels, which is related to the quality analysis results of the accumulated waveforms. Therefore, considering the algorithm efficiency and ranging performance, only a few high-quality channels should be selected to retrieve the position distribution of targets, such as 20 channels here. The channels participating in the multichannel accumulation process are flexibly selected, suitable for different band combinations and various scenarios. Furthermore, the quantitative results, i.e., noise standard deviation (N_{std}), SNR, and ranging precision (E_p) of the three targets are listed in Table III.

Table III displays the comparison results of signal quality and ranging precision obtained from single-channel (730 nm) waveforms and multichannel (20 channels) accumulated waveforms. The spectral channel (730 nm) with the best signal quality among the multiple channels is taken as the reference benchmark to illustrate the advantages of the MIWD method. Compared with the results of the optimal channel, both of the two accumulation methods reduce the background noise level by over three times, increase the SNR by nearly 10 dB, and increase the ranging precision by over two times. The results reveal that the ranging performance of the MIWD method can be considerably improved through multichannel weighted accumulation compared with the traditional SWWD method given that the data quality of the waveform is improved during the process. In order to realize real-time and efficient laser ranging, the waveform processing algorithm is implemented on field-programmable gate array (FPGA) with the powerful ability of parallel compute, processing waveform in an online manner. To further evaluate the ability of the proposed method to retrieve spectral information in long-distance detection, the random distribution of the reflectance of leaves under different health conditions measured by the FWHSL system is presented in Fig. 7.

Fig. 7 exhibits the reflectance curves of the green and yellow leaves covering 500–800 nm. Generally, the green leaf shows significantly lower reflectance at 650–700 nm compared with

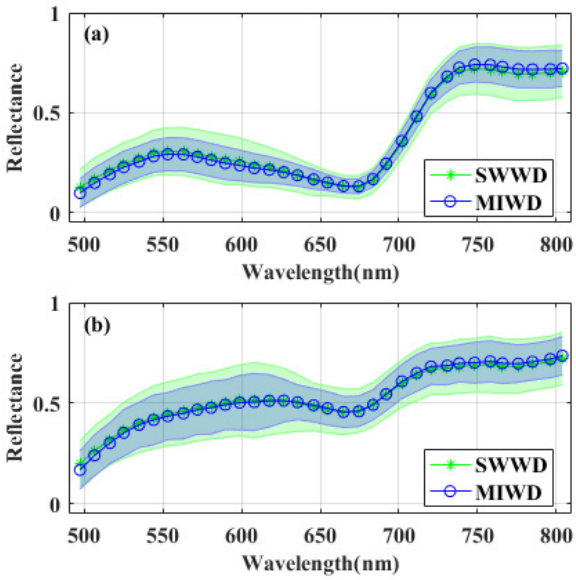


Fig. 7. Spectral measurements of (a) the green and (b) yellow leaves through the FWHSL system using SWWD and MIWD, respectively. The colored dotted splines represent the mean values of the spectral distribution of the green and yellow leaves obtained using SWWD and MIWD, and the corresponding shaded area represents the range of numerical variation.

the yellow leaf due to the absorption peak of chlorophyll around 675 nm. This means that the FWHSL system has the ability to monitor the health condition of vegetation. The mean values of spectral distribution obtained using the two methods are approximately identical. Particularly, the shaded area presents the uncertainty of the measured reflectance. Compared with the SWWD method, the proposed method combines multiple channels to extract the position of the waveform echoes more accurately; hence, it is less susceptible to the background noise during the waveform fitting process, thereby reducing the uncertainty of reflectance as shown in the shaded area.

C. Overlapping Echo Decomposition With MIWD

To further explore the ability to identify adjacent targets at the long-distance detection, waveform data with overlapping echo components were collected from two standard whiteboards at around 518 m. The distance between these two targets was changed from 40 to 120 cm with a step size of 20 cm by moving the second plane away from the first plane. As discussed in Section IV-B, the multichannel accumulation is carried out in order to enhance the hyperspectral waveforms with multiple echo components. Given that the

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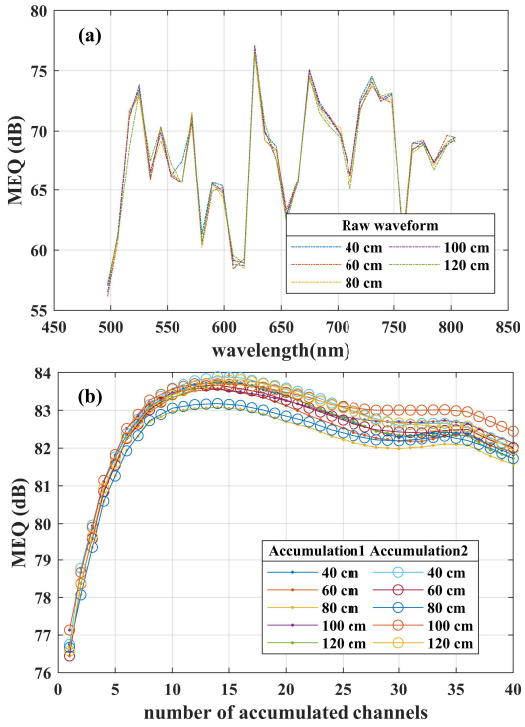


Fig. 8. Data quality analysis results of (a) raw waveforms at each spectral channel and (b) accumulated waveforms.

recorded waveforms contain multiple peaks, the MEQ index introduced in Section III-B is deployed to evaluate the signal quality instead of SNR, which is only related to the maximum value of the echo waveform. Fig. 8 demonstrates the significant improvement of data quality using the MIWD method with the accumulated channels increasing to 14. This means that the waveform accumulation achieved the best signal enhancement effect when employing 14 channels with the largest MEQ values. Compared with the channel (626.8 nm) with the largest MEQ value, the accumulation of the 14 selected channels improves the signal quality greatly from 77 to 83 dB.

Fig. 8 demonstrates the corresponding raw waveform data with different degrees of overlap. For instance, Fig. 8(a) displays the hyperspectral waveform echoes with an interval of 40 cm. The echoes vary at different channels, and only part of the channels can vaguely show two echo components. This implies that compared with a single channel, multichannel waveforms can provide richer information and potentially retrieve more details in a long-distance detection. In order to clearly show the overlapping effect of the adjacent echoes, Fig. 9(b)–(f) presents the intensity distribution of the waveforms in the two dimensions of wavelength and distance, corresponding to the interval of 40–120 cm.

When the interval of the two targets is 40 or 60 cm as marked by the red frame in Fig. 9(b) and (c), the two echoes are extremely close to each other and appear as a single widened echo, thereby, it is difficult to distinguish the two echoes spatially for each spectral channel. Although the second echoes can be roughly identified when the interval of the two targets is 80 or 100 cm as marked by the yellow frame in Fig. 9(d) and (e), they are very likely to be judged as noise and ignored due to the weak response in each channel. In order to explore the ability to extract the spatial information of multiple targets in detail, we carried out the MIWD method on waveform data using multiple channels. Fig. 10 displays the decomposition results of the 14-channel accumulated waveforms according to the analysis in Fig. 8, compared with the results of the optimal spectral channel, 626.8 nm.

Fig. 10 exhibits that more details can be retrieved from the multichannel accumulated waveforms. Specifically, when the interval of the two targets is 40 or 60 cm [Fig. 10(a), (b), (f), and (g)], only the accumulated waveforms extracted two components. This indicates that single-channel waveforms may bring tremendous uncertainty to the waveform decomposition, resulting in component loss or inaccurate parameter extraction. Benefiting from the abundant spectral information with multiple channels, the two waveform components could be decomposed from the accumulated waveforms, which is more reliable and offers a considerable improvement over the single channel. Moreover, 1000 pulse echoes were randomly selected from each set of experimental data and processed in order to evaluate the waveform fitting accuracy of the proposed method, and the results of R^2, RMSE as well as MEQ are listed in Table IV.

Table IV verifies that the MIWD method has great advantages in improving the overall data quality and fitting accuracy at the long-distance detection. The MEQ values indicate that the signal quality was greatly improved through the multichannel accumulation process. The R^2 and RMSE values suggest that the decomposition results derived from multiple channels exhibit a comparable overall fitting accuracy as the single channel. However, the overall fitting accuracy as measured by R^2 and RMSE values offers only a general view of algorithmic performance when approximating the raw waveform data. In order to further evaluate the ability of the detailed information acquisition through the waveform decomposition method, the success rate of extracting two waveform components, the measured interval values, and RNDE are listed in Table V.

Table V illustrates the improvement in discovering overlapping echo components using the MIWD method from the perspective of statistical probability. The proposed method has a relatively higher success rate in extracting waveform components than only using the single channel especially when the interval is 40 or 60 cm. The MIWD method could distinguish targets spaced 40 cm apart with a success rate of over 40%, whereas only 26% of waveforms with two components spaced 60 cm apart could be separated from the single channel. Although the probability of detecting two components in a single channel exceeds 70% when the interval is over 80 cm, the RNDE values (MIWD with A2: 0.0291 and SWWD with 626.8 nm: 0.0896) indicate that the accuracy of the measured interval is greatly improved using the MIWD method. Therefore, the proposed method exhibits great advantages in extracting detailed information of multiple targets by using multiple wavelength combinations in a long-distance detection.

V. Discussion

This study is the first attempt to retrieve the spatial and spectral information of long-distance (>500 m, which is

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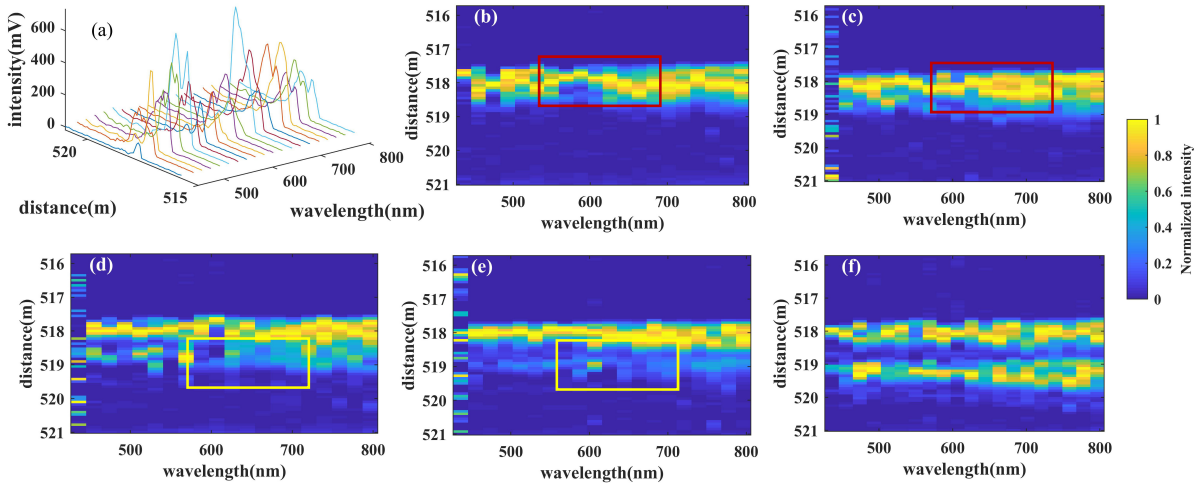


Fig. 9. Schematic of raw waveform data with overlapping echoes. (a) Hyperspectral waveform echoes with an interval of 40 cm. (b)–(f) Distribution of the normalized intensity of the echo waveforms over the wavelength and distance, corresponding to the interval of 40–120 cm.

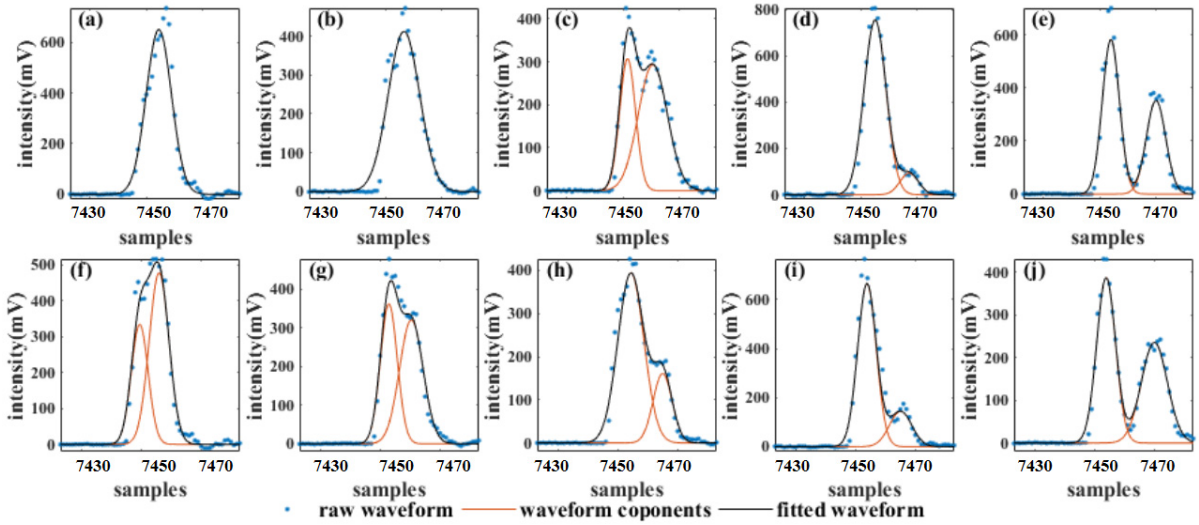


Fig. 10. Decomposition results of the waveforms at 626.8 nm and the accumulated waveforms of 14 channels. (a)–(e) Decomposition results of the single-channel (626.8 nm) waveforms with an interval of 40, 60, 80, 100, and 120 cm. (f)–(j) Decomposition results of the accumulated waveforms of 14 channels with an interval of 40, 60, 80, 100, and 120 cm.

TABLE IV

DATA QUALITY AND FITTING ACCURACY OF DECOMPOSITION RESULTS BASED ON SINGLE CHANNEL AND MULTIPLE CHANNELS

True interval (cm)	MEQ (dB)			R^2			RMSE		
	S	A1	A2	S	A1	A2	S	A1	A2
43	76.75	83.74	83.89	0.95	0.95	0.97	5.97	5.74	5.94
60	76.44	83.56	83.66	0.97	0.96	0.94	3.53	3.58	3.55
80	76.67	83.08	83.18	0.95	0.95	0.93	3.80	3.07	3.66
100	77.13	83.57	83.70	0.97	0.95	0.94	3.93	3.10	3.05
120	76.57	83.72	83.81	0.95	0.94	0.94	3.93	3.65	3.98

S: Single channel (626.8 nm), A1: Accumulation1, A2: Accumulation2.

a suitable distance for the airborne platform) targets using FWHSL measurements. Previous studies have focused on the spectral detection capability by increasing receiving spectral channels, which have demonstrated great advantages of the FWHSL systems in obtaining vertical spectral and biochemical distribution of vegetation [18], [21]. Despite this, the relevant experiments were carried out in laboratory circumstances from a few meters to a maximum effective range of roughly 20 m [26], [28], which limits the application of the FWHSL systems in large-scale long-distance detection. In this study, we have fully considered the characteristics of hyperspectral waveform data in the long-distance detection, and proposed a

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TABLE V
COMPONENT RELATED RESULTS BASED ON SINGLE CHANNEL AND MULTIPLE CHANNELS

True interval (cm)	Success rate (%)			Measured interval (cm)			RNDE		
	S	A1	A2	S	A1	A2	S	A1	A2
40	3	45	42	63.25	44.58	47.83	0.5813	0.1145	0.1958
60	47	66	73	70.58	55.42	56.17	0.1763	0.0763	0.0638
80	75	84	83	72.83	75.58	77.67	0.0896	0.0553	0.0291
100	85	89	91	110.50	98.17	97.83	0.1050	0.0183	0.0217
120	92	97	97	127.93	119.50	123.50	0.0661	0.0042	0.0125

S: Single channel (626.8 nm), A1: Accumulation1, A2: Accumulation2.

complete processing flow, MIWD, to obtain the accurate distance and spectral information of targets. First, the waveform echoes theoretically have spatial consistency on each channel, while the receiving channels are not strictly synchronized in time. Given that each receiving channel has collected the transmitted pulse and the echo waveform, we calibrated the time deviation between channels in the laboratory in advance according to the peak positions of the recorded waveforms, regardless of atmospheric influence.

Second, the targets may have low reflectivity in certain bands and the supercontinuum laser has extremely low output energy near 450 nm, leading to poor signal quality in multiple receiving channels [28] especially for the long-distance detection. Previous studies [19], [41] have confirmed that the poor data quality greatly affects the waveform decomposition accuracy. Considering that multipulse accumulation can effectively improve the SNR [34], we designed a strategy of multichannel accumulation to improve the overall signal quality to a great extent.

Third, this study analyzed the data quality and ranging precision under different channel accumulations and provided the optimal number of accumulated channels for each experiment. The results indicated that when retrieving the position distribution of targets, adding too many channels with a relatively low SNR was not conducive to data processing. Therefore, the best several channels were input into the accumulation process to retrieve the accurate position information, and then the spectral-related information of the waveform components in all the spectral channels is retrieved. The channels participating in the accumulation should be flexibly set according to the detection scenario and the receiving bands.

Finally, this study assumed that the echo waveform conforms to the Gaussian distribution and further proposed multichannel interconnection decomposition method. Since the multichannel interconnection mode is designed to improve the quality of waveform data and thus the accuracy of waveform decomposition, it can be extended to other single-channel waveform decomposition methods, including mathematical simulation and deconvolution algorithms. For the mathematical simulation methods, in addition to Gaussian distribution, other suitable kernel functions should be selected according to the transmitted pulse, such as logarithmic and adaptive Gaussian functions [34]. As for the deconvolution algorithms, the Richardson–Lucky (RL) algorithm performs better in detecting overlapping subwaveforms from received waveforms, but is an ill-conditioned problem and susceptible to noise [5]. Moreover, the computational complexity of the RL algorithm makes it more time-consuming than the Gaussian decomposition, especially for massive LiDAR data.

Further experiments under extensive outdoor environments should be conducted with the FWHSL system to validate the performance of the FWHSL system in retrieving multiple types of targets in the airborne scene. The proposed method should also be extended to other FWHSL systems. The FWHSL system, with detectors covering a large wavelength range and a supercontinuum laser with high-output energy especially for near 450 nm may reveal more details about targets.

VI. CONCLUSION

This article proposed the MIWD method to explore the ability of detecting long-range targets using the FWHSL data. The performance of the proposed method is evaluated in terms of ranging, spectral detection, and adjacent target separation using the FWHSL measurements. The spectral channel with the best signal quality among the multiple channels was taken as the reference benchmark to illustrate the advantages of the MIWD method. Compared with the results of the optimal channel, MIWD significantly improved the ranging precision by over two times. The accurate extraction of the target position distribution then greatly reduced the uncertainty of the spectral information retrieval. MIWD also showed considerable improvement in identifying adjacent distance, with the success rate of 60-cm spacing increased from 47% to 73%. Ultimately, the proposed method effectively utilized FWHSL data to significantly improve the waveform decomposition accuracy, ranging precision, and overlapping component separation in the long-distance detection. The proposed method will combine other kernel functions or deconvolution algorithms to improve the universality of the algorithm. It will be integrated into FPGA to realize online processing to improve computing efficiency, and the FWHSL system is planned to be carried on an airborne platform on the basis of the performance test in this study.

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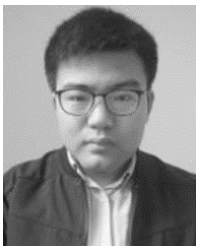
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